

Causal tomography: certifying manifoldlikeness of a finite causal order from two-dimensional shadows

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preprint, not peer reviewed, version 2026-07-02. This paper is self-contained. The theorems (Sections 2–4) are elementary and proved in-paper given the named imports (Gallai’s transitive-orientation theorem, Logan’s finite-angle uncertainty principle); the numerical demonstrations are float64 measurement landscapes with fixed seed `default_rng(20260702)` and measured-and-disclosed tolerances, and the repository ships the complete demonstration code and canonical outputs (receipts `i1–i3`, `k1–k3`, `l1–l3`, `m1` in `code/`, outputs in `receipts/`). Claims are graded in place: **[THEOREM]** = proved here; **[DEMONSTRATED]** = measured with shipped code at the stated scale; verdicts that fall short of a gate are labeled **inconclusive**, not certified.

Abstract

Whether a finite partial order is “manifoldlike” — approximable by a uniform-density sprinkling of a Lorentzian spacetime — is the causal-set form of the Hauptvermutung, and above 1+1 dimensions it collides with an order-theoretic wall: Minkowski dimension detaches from Dushnik–Miller order dimension, and recognition of the relevant order classes is believed hard. We do not attack the decision problem. We split **verification from search**: we construct a polynomial-time-verifiable *certificate* of higher-dimensional volume-faithfulness — possessing one is proof, up to an exactly characterized budget — and treat finding one as heuristic search whose hardness is expected and priced. Four results. **(1) A 2D engine [THEOREM]**. For orders of dimension ≥ 2 we prove a per-order, deterministic, quantitative characterization: P admits a volume-faithful embedding into the M^2 diamond with anchored-box defect $\leq \epsilon$ iff $\dim(P) \geq 2$ and the *canonical rank embedding* — built order-intrinsically from the essentially unique realizer (Gallai) — has star discrepancy $D^* \leq 3\epsilon + 1/n$ (backward direction exact). A relabeling lemma shows the residual realizer freedom (modules) moves the embedded point set not at all; the one exception class (internal prime flips) is closed by census and priced below 10^{-6} where it occurs. **(2) Causal tomography [THEOREM + DEMONSTRATED]**. The order of M^{1+d-1} is the intersection of its null-coordinate total orders over the celestial sphere; each antipodal direction pair projects the order to a 2D **shadow** whose target density is computable in closed form (the chord law $\sqrt{(1-|t|)^2 - s^2}$ in 2+1; the disk-section law $(1-|t|)^2 - s^2$ in 3+1). Scoring shadows with a weighted discrepancy against the target copula, plus a **harmonic condition** — direction profiles of a faithful order are degree-1 (Fourier / spherical) harmonics whose coefficients *are* the coordinates — yields certificate conditions that reconstruct coordinates at correlation 0.9994 per axis in 2+1 and 0.9999 per axis in 3+1 (oracle-family), including, to our knowledge, the first reconstruction of the **celestial sphere of a causal set from inversion counts alone** (S^1 : radius CV 0.009; S^2 : radius CV 0.008, angle correlation 0.9995). **(3) The ghost budget [THEOREM-shaped + DEMONSTRATED]**. The certificate’s soundness

is scoped exactly: every marginal-preserving rearrangement is either negligible or a spectral density modulation; a mode invisible to K directions must have frequency $q \approx 1/\sin(\pi/2K)$ (Logan), and any such mode moves every anchored-box count by at most a/q — so the certificate controls the discrepancy metric at all scales, while high-frequency spectral non-uniformity below that amplitude is provably outside *any* finite- K discrepancy certificate. Both halves are demonstrated: a constructed $q = 120$ ghost (genuinely non-uniform, matched-filter $z = 15.3$) passes the full certificate; the same amplitude at $q = 6$ is caught at over twice the calibrated floor. **(4) An intrinsic finder [DEMONSTRATED]**. A fully order-intrinsic search pipeline — a past/future-count temporal seed, a chain-fleet transverse embedding, an order-only scale calibration, and a *tomographic bootstrap* that uses the reconstruction operator as its own polish — collapses the search gap from $6\times$ the certificate floor (annealing from cold) to $1.5\times$ at $N = 400$ and to $1.21\times$ at $N = 1000$ in $2+1$ — the latter *inside* the certificate gate for the shadow condition — and runs end-to-end in $3+1$ at $1.50\times$ (verdict: inconclusive, at exact parity with the $2+1$ pilot’s first stage). The same pipeline *refuses* two non-manifoldlike adversary classes. Verification is solved at demonstration scale with its soundness budget stated exactly; search is measurably close; certified end-to-end $3+1$ reconstruction is the stated open problem.

1. The problem and the reframe

A causal set is a locally finite partial order proposed as the deep structure of spacetime [1]. Its central mathematical problem is manifoldlikeness: which orders arise (with high probability, or approximately) as Poisson sprinklings of a Lorentzian manifold, with the order given by the causal relation [2, 3]? By Malament’s theorem the causal order of a distinguishing spacetime determines its conformal geometry [4], and sprinkling density supplies the volume factor — “order + number = geometry”. The verification form of the question, per order: *given one finite order P , decide — and quantify — whether it embeds in a spacetime region at uniform density*.

In $1+1$ dimensions the order theory cooperates: embeddability in M^2 is exactly Dushnik–Miller dimension -2 , recognizable in polynomial time [5, 6]. Above that the wall is real: the realizer route collapses (spacelike configurations of M^{1+2} induce circle containment orders, for whose recognition no polynomial algorithm is known and which are believed hard [7]), and the field’s practical instruments — dimension estimators [8, 9], action-based tests [10], embedding procedures [11] — are either necessary conditions only, or procedures without theorems.

We therefore stop attacking symmetric decision and split the problem NP-style. A **certificate** of d -dimensional volume-faithfulness is a witness structure W such that *(T1)* verifying W is polynomial-time and yields a constructive embedding with a defect bound, and *(T2)* every faithful order possesses a W (its true geometric structure). Finding W is then honest heuristic territory: worst-case hard, as it must be — but *any* success is a proof, not a score. Sections 2–3 build the certificate; Section 4 scopes its soundness exactly; Section 5 attacks the search face; Section 6 runs the $3+1$ stack. Throughout, D^* denotes anchored-box (star) discrepancy of a planar point set, and all gates, floors, and tolerances are calibrated and disclosed in the shipped receipts.

2. The two-dimensional engine

Everything downstream scores 2D projections, so the 2D case must be solved *per order*, deterministically and quantitatively. Prior results in this regime are typicality statements — the uniform measure on 2D orders concentrates on continuum-like orders [12], and random 2D orders are sprinklings of the diamond in distribution [13] — or embedding procedures without canonicity. We prove:

Theorem 2.1 (canonical realizer). A 2D order whose incomparability graph is prime (in the modular-decomposition sense) has exactly two transitive orientations (Gallai [14]), hence exactly one realizer pair (L, L') up to the parity swap $u \leftrightarrow v$. For non-prime orders the residual freedom is exactly the module freedom, and the **relabeling lemma** disposes of almost all of it: series modules admit no freedom, and a parallel block’s quotient enters L in one order and L' reversed, so reorienting it permutes which element carries which coordinate pair — the embedded point *set*, hence D^* , is exactly invariant. Only *internal prime-node* flips can move the point set. (*Demonstrated: bit-level invariance on a population carrying 105 modules at $N = 396$; the internal-prime residue closed by census — zero point-moving modules on sprinkling and lattice populations; on the one population where the freedom is real, 143 point-moving blocks are exhibited and priced at worst $|\Delta D^*| < 10$. Receipts i1, i3.*)

Theorem 2.2 (volume gauge). A 2D order’s embedding freedom is exactly a pair of monotone reparametrizations of its two null coordinates — the discrete conformal freedom. Demanding uniform marginals fixes both: the **canonical rank embedding** (each null coordinate replaced by its normalized rank) is the volume-normalized representative of the order’s conformal class, realizing the order exactly with exactly uniform marginals.

Theorem 2.3 (characterization). For a 2D order P on n elements, both directions quantitative: (*forward*) if P admits a volume-faithful embedding with anchored-box defect ≤ 2 , then $\dim(P) \leq 2$ and the canonical rank embedding has star discrepancy $D^* \leq 3 + 1/n$; (*backward, exact*) if the canonical rank embedding has $D^* = 2$, then P is $\sqrt{2}$ -volume-faithful — the rank embedding itself is such an embedding, and its defect *is* its D^* . Forward proof: any $\sqrt{2}$ -faithful embedding’s coordinate orders are the canonical realizer up to parity and relabeling-invariant module freedom, and monotone reparametrization to uniform marginals moves anchored-box mass by ≤ 2 , plus the $1/n$ rank discretization.

The theorem is constructive and the pipeline is polynomial (dimension-2 recognition and modular decomposition are classical polynomial problems [6, 15]; exact D^* is $O(n^2 \log n)$ here). The showcase measurement: monotone distortions $(u^2, \sqrt{v})$ of a faithful embedding leave the order bit-identical while inflating the raw point-set discrepancy $0.053 \rightarrow 0.301$; the canonical rank embedding recovers the volume gauge *from the order alone* at $D^* = 0.036$ (receipt i2). Two honest boundaries, both measured: a bounded blow-up adversary (clusters of c co-located points) is flagged at finite N as a $\sqrt{c} D^*$ -inflation *diagnostic* (measured $2.45\times$ against the predicted $\sqrt{6} = 2.45$) but is asymptotically volume-faithful — its outright rejection belongs to a separate *statistical* layer (interval-count dispersion; measured kill at $z = 28$); and volume-faithful Poisson-typical — a jittered lattice rightly passes the geometric characterization and fails the statistical layer ($z = 5.2$ sub-ensemble dispersion). The characterization is a theorem about geometry; Poisson typicality is a model question deliberately split off.

3. Causal tomography in 2+1

Theorem 3.1 (shadows). For P the order of a finite point set in $M^{1+(d-1)}$: for every spatial unit vector $\mathbf{m}$, the projection $\mathbf{x} \mapsto (\mathbf{t}, \mathbf{m} \cdot \mathbf{x})$ maps P to a 2D dominance order that coarsens P , with null coordinates the functionals $u_{\pm \mathbf{m}} = \mathbf{t} \pm \mathbf{m} \cdot \mathbf{x}$; and P is exactly the intersection of its shadow orders over the continuum celestial sphere (per pair, the supremum is achieved at $\mathbf{m} = \Delta \mathbf{x} / |\Delta \mathbf{x}|$). *(The coarsening half is exact in-receipt; the intersection identity’s finite- K face is measured — a K -direction grid intersection converges to P from above, extra relations 32/11/1 of 17,291 comparable pairs at $K = 16/32/64$. Receipt k1.)*

A faithful diamond sprinkling’s shadow is not uniform: its density is the **chord law** $(\mathbf{t}, \mathbf{s}) \mapsto \sqrt{((1-|\mathbf{t}|)^2 - \mathbf{s}^2)}$ (verified at total variation 0.023 against a 200k-point projection). The Section 2 certifier generalizes in one move: score each shadow’s point set against the *target copula* (the chord law pushed through its own quantile transforms — which also absorbs any monotone gauge freedom per direction) with a weighted star discrepancy $\mathbf{wD}^*$, floor-calibrated by same-law subsamples. Measured at $N = 400$: true shadows sit at $1.40\times$ the two-sample floor (gate $1.5\times$), random-extension garbage at $9.4\times$; the fine print is printed — the chord copula’s departure from uniformity is small (0.040 asymptotically), so at this N the weighting earns its keep in defect attribution and at scale, not as a dramatic finite- N kill.

The harmonic condition. For true Minkowski coordinates, $u_{\pm}(\mathbf{x}) = \mathbf{t} \pm \mathbf{x} \cos \theta \pm \mathbf{y} \sin \theta$: the direction profile of every point is *exactly* a degree-1 Fourier polynomial in θ , and the coefficients $(\mathbf{a}, -\mathbf{a}, -\mathbf{b})$ are $(\mathbf{t}, \mathbf{x}, \mathbf{y})$. The certificate condition demands quantile-gauged profiles (each $u_{\pm}$ recovered intrinsically from ranks through the target marginal) be degree-1 concentrated. Measured: RMS residual 0.020; coordinate reconstruction at correlation 0.9994 per axis; the rebuilt point set reproduces the original order on 99.67% of pairs and matches anchored-box counts to a mean absolute defect of 2.1 elements. Faithfulness transfers to the reconstruction with no closed-form volume formula anywhere.

The celestial circle from inversion counts. Classical multidimensional scaling on pairwise Kendall (inversion) distances between the 16 null-direction extensions recovers S^1 — radius CV 0.009, cyclic adjacency 16/16 — from order data alone, *given the extension family*. Finding the family intrinsically is the search face (Section 5).

Two measured honesty points. The certifier has a *plateau*: 400 adjacent-swap corruptions of the extensions leave the score at the floor, and corrupting all 16 directions still reconstructs coordinates at > 0.999 per axis — certificate-grade families are interchangeable for reconstruction, which is aligned with the certificate’s semantics (any certified witness yields the geometry). And annealing from a random start stalls at $6.0\times$ the floor after 12k local moves — the hardness face, exactly where the reframe put it. *(Receipts k2, k3 — the latter also demonstrates the chain-fleet transverse principle of Section 5: greedy chain extraction, cross-chain comparability coupling, MDS recovers the transverse plane at Procrustes 0.993, shuffled-coupling control 0.19.)*

4. Certificate v2 and the ghost budget

Certificate v2. For a d -dimensional claim over direction set $\mathbf{m}_k$: (i) every antipodal shadow scores within a max-calibrated same-law floor (gate $1.3\times$ the floor in 2+1, $1.35\times$ in 3+1, replicate-

calibrated); (ii) harmonic residual < 0.05 ; (iii) the harmonic reconstruction *realizes the input order* at pair agreement 0.98 . Verdict bands, fixed in advance: $< 1.3\times$ certified; $1.3\text{--}2\times$ inconclusive; $> 2\times$ refused. Condition (iii) is what makes soundness discussable — the certificate asserts *this order embeds as this point set*, not merely *some projections look right* — and it is measurably load-bearing: a **separator adversary** (an extension family lifted from a *different* faithful configuration) passes (i) at 0.040 and (ii) at 0.012 and fails (iii) at agreement 0.794 (receipt 11).

The soundness budget, closed into a loop. Discrete tomography’s classical ghosts — switching components [16, 17] — dichotomize here: small-support rearrangements leave the configuration faithful up to negligible defect (the certificate rightly still passes — the input is simply another faithful order), while extended switching patterns *are* density modulations. So the true adversary is spectral: $1 + \mathbf{a} \cdot \cos(\mathbf{q} \cdot \mathbf{x})$. Two elementary facts pin its fate. (a) Its shadow marginal in direction $\mathbf{n}_k$ is fiber-averaged, leaving relative leakage $\mathbf{a}/(\mathbf{q} \cdot \sin(\hat{\mathbf{q}}, \mathbf{n}_k))$ — so invisibility to all K directions forces $\mathbf{q} = 1/\sin(1/2K)$: Logan’s finite-angle uncertainty principle [18] in our setting. (b) Any mode’s contribution to any anchored-box count — in a shadow or in the bulk — is at most $\mathbf{a}/\mathbf{q}$, because the CDF of an oscillation is flat to $1/\mathbf{q}$. Together: **certificate-invisible ghosts cannot move the discrepancy metric anywhere**; what they can move is spectral density below wavelength $2/\mathbf{q}$. Demonstrated on all four legs (receipt 11): a $\mathbf{q} = 120$ ghost at $\mathbf{a} = 0.6$ is genuinely present (matched filter $\mathbf{z} = 15.3$); it passes the full $\mathbf{v}_2$ certificate at $K = 8$ and its shadows at $K = 16$ (more directions do not catch it, exactly as (b) predicts) while bulk box discrepancy stays within its disclosed gate ($0.027 \rightarrow 0.039$); and the same amplitude at $\mathbf{q} = 6$ is caught at over $2\times$ the floor. **The guarantee, final form:** certificate $\mathbf{v}_2$ implies an order-realizing embedding with anchored-box volume defect controlled at all scales; spectral non-uniformity above frequency K *can be* invisible whenever its amplitude sits below the CDF floor — a budget that is metric-intrinsic (it binds any discrepancy-type certificate), amplitude-conditioned, and extendable by an explicit spectral axis (matched filters per frequency band) for callers who need it.

5. The intrinsic finder

The search face closes to near-certificate range with a fully order-intrinsic pipeline, no annealing: (1) temporal seed $\mathbf{t} = |\text{past}(\mathbf{x})|^{1/d} - |\text{future}(\mathbf{x})|^{1/d}$ with d the spacetime dimension (cone-volume counting: $1/3$ in $2+1$, $1/4$ in $3+1$); (2) transverse seed from a greedy chain fleet — extract long chains, form the cross-chain comparability coupling $\mathbf{w}(\mathbf{c}, \mathbf{c})$, embed chains by MDS, extend to all elements by coupling-weighted averaging; (3) an order-only scale calibration (the MDS scale is arbitrary against $\mathbf{t}$; a 12-point geometric scan on the certificate score fixes it — omitting this step scores $2.5\times$, receipted); (4) the **tomographic bootstrap**: quantile-gauge each direction’s profile through the target marginal, harmonically fuse into refined coordinates, iterate two to three times — the reconstruction operator as its own polish.

Measured in $2+1$ (receipts 12, m1): at $N = 400$, $\max \mathbf{wD}^* = 0.048$ vs floor 0.033 — $1.5\times$, inconclusive band, with condition (iii) at 0.953 ; at $N = 1000$ ($K = 12$), 0.0228 vs 0.0188 — **$1.21\times$, inside the $1.3\times$ gate for condition (i)** — with (iii) at 0.9648 , still short of its 0.98 gate, so the overall verdict remains inconclusive and the residual is (iii)-only at this scale. The $1.5\times \rightarrow 1.21\times$ move measures the condition-(i) gap as finite-size, not structural. The refusal twin holds: the same pipeline refuses a Kleitman–Rothschild-type layered order (an explicit

degenerate-fleet refusal path — such orders have no long-chain structure to seed from) and scores the cluster adversary at 2.2× (refused band).

6. Three-plus-one dimensions

The stack transfers with no structural change (receipt 13; $N = 900$, 12 Fibonacci-spiral directions plus antipodes). The shadow target becomes the **disk-section law** $(1 - |\mathbf{t}|)^2 - \mathbf{s}^2$ (TV 0.013); all 12 true shadows pass at the max-calibrated floor (gate 1.35×, 10-replicate calibration, pass margin +0.10 floors — small, and said so; fairly quantile-mapped garbage scores 4.2× above the true band); the harmonic condition becomes degree-1 *spherical* harmonics — profiles affine in $n - \mathbf{S}^2$, residual 0.012 — whose four coefficients reconstruct $(\mathbf{t}, \mathbf{x}, \mathbf{y}, \mathbf{z})$ at 0.9999/0.9999/0.9998/0.9999 with the rebuilt order agreeing on 99.88% of pairs. The flagship: MDS on pairwise inversion distances between the 24 signed-direction extensions yields a 3D shell of radius CV 0.008, a decisively 3-dimensional spectrum (fourth-to-third eigenvalue ratio 0.019, gated on $|\mathbf{t}|/|\mathbf{s}|$), and pairwise angles matching the true celestial angles at correlation 0.9995 — to our knowledge the first reconstruction of the celestial sphere of a causal set from inversion counts. These families are **oracle-derived** (built from the true coordinates, as a verification of the certificate, not a search result), and we say so.

The intrinsic pipeline of Section 5 then runs end-to-end in 3+1 on the same configuration (receipt m1): seed quality $\mathbf{t}$ correlation 0.961 and transverse Procrustes 0.888 from $M = 26$ chains (the 4D chains are shorter and the transverse space larger — the seed bars are deliberately lower and disclosed); after calibration and bootstrap, $\max \mathbf{wD}^* = 0.0372$ vs floor 0.0247 — **1.50×, condition (iii) 0.970: inconclusive, at exact parity with the 2+1 pilot’s own first stage**. Held-out reconstruction: $\mathbf{t}$ correlation 0.974; spatial correlation 0.916 after Procrustes alignment (the recovered spatial frame is defined only up to global rotation, so per-axis spatial correlations are undefined under the certificate’s own symmetry). Given Section 5’s scale lesson, the larger- N 3+1 run is a compute question, not a structural one — but the certified end-to-end 3+1 claim is **not made here**.

7. Limitations and open problems

All demonstrations are flat-diamond sprinklings at $N = 1200$; curved backgrounds, boundaries, and non-diamond regions are untouched. The ghost budget’s two legs are proved at the physics-paper level (fiber-averaging leakage; CDF flatness) with Logan cited for the principle; sharp constants and the finite- N noise-vs-leakage interplay are measured regimes, not theorems. The finder’s refusal behavior is measured on two adversary classes, not proved. T1/T2 as stated in Section 1 are demonstrated constructively at pilot scale; their journal-grade quantitative forms (defect-propagation constants, the $\mathbf{S}^2$ version of the harmonic argument) are open. The named open problems, in order of expected yield: (1) the certified end-to-end 3+1 reconstruction — close condition (iii)’s gap (0.965/0.970 vs 0.98) and run the 3+1 finder at larger N ; (2) the sharp ghost-budget constants; (3) the statistical (Poisson-typicality) layer as a characterization rather than a battery of tests; (4) curved/conformally-flat extensions of the 2D engine.

8. Code and reproducibility

All code is in `code/` and all canonical outputs in `receipts/` of this repository. Every number quoted above appears in a receipt output. The scripts are self-contained (NumPy only), run in minutes on a laptop, and print explicit `[PASS]/[FAIL]` gates with their tolerances; a script exits nonzero if any check fails.

- `i1_canonical_realizer.py` — Theorem 2.1: realizer canonicity, the relabeling lemma’s bit-level invariance exhibit, Gallai verified live by small- n enumeration. Receipt: `i1_canonical_realizer.out`.
- `i2_characterization.py` — Theorems 2.2–2.3: the rank embedding, the $(u^2, \sqrt{v})$ distortion showcase ($0.301 \rightarrow 0.036$), the blow-up diagnostic ($2.45\times$ vs $\sqrt{6}$), the statistical-layer split (lattice, $z = 5.2$). Receipt: `i2_characterization.out`.
- `i3_internal_prime_census.py` — the internal-prime census closing Theorem 2.1’s residue: absence on sprinkling/lattice populations; 143 point-moving blocks exhibited and priced < 10 on the cluster population. Receipt: `i3_internal_prime_census.out`.
- `k1_shadow_engine.py` — Theorem 3.1’s measured faces, the chord law (TV 0.023), the weighted certifier with floor calibration, the harmonic reconstruction ($0.9994/\text{axis}$, 99.67% order agreement). Receipt: `k1_shadow_engine.out`.
- `k2_shadow_search.py` — the celestial circle (CV 0.009, adjacency 16/16), the plateau and all-directions-corrupted reconstruction ($> 0.999/\text{axis}$), the cold-start stall ($6.0\times$). Receipt: `k2_shadow_search.out`.
- `k3_decoration_probe.py` — the chain-fleet transverse principle (Procrustes 0.993; shuffled control 0.19). Receipt: `k3_decoration_probe.out`.
- `l1_ghost_budget.py` — certificate v2 jointly verified; the $q = 120$ ghost’s four legs; the $q = 6$ control; the separator adversary (condition (iii) load-bearing). Receipt: `l1_ghost_budget.out`.
- `l2_intrinsic_finder.py` — the Section 5 pipeline at $N = 400$ ($1.5\times$; naive-seed $2.5\times$; refusals). Receipt: `l2_intrinsic_finder.out`.
- `l3_four_dimensional.py` — the 3+1 stack (disk-section law; 12/12 shadows; spherical harmonics $0.9999/\text{axis}$; the celestial S^2). Receipt: `l3_four_dimensional.out`.
- `m1_intrinsic_4d.py` — the 2+1 scale test ($1.21\times$ at $N = 1000$) and the intrinsic 3+1 run ($1.50\times$, (iii) 0.970). Receipt: `m1_intrinsic_4d.out`.

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